NIAID BLUEPRINT WEBINAR COMPANION · 6 NOVEMBER 2026

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> **Level 1 view — publish or update served files.** Only Level 1 items are included. These are file-shaped actions. Publishing a file may still require an administrator. Higher-level configuration and services appear in the complete checklist. Showing 14 items.

Repository checklist: making your repository more accessible to SI

Companion to the 6 November 2026 NIAID Blueprint webinar *Making Your Repository More Accessible to SI* and its supplementary materials.

This file is the **only content authority** for the checklist. Generated HTML and PDF views are renderings of this document. Edit here, then rebuild.

This companion is a practical list of actions. It is **not** a certification, audit rubric, maturity model, or compliance score. Finishing more items, or items at a higher level number, does not make a repository more trustworthy, more FAIR, or more "SI-ready." Levels describe the kind of access and maintenance an action typically needs, not quality.

The webinar focuses on assets a repository exposes to agents and workflows—configuration, files and metadata, APIs and MCP services, and semantic-layer components. It does not assess whether particular datasets are suitable for model training or inference.

Informational only; not legal advice.

How the levels work

Levels record **capability and access**, not a ranking of repositories.

Level	What it usually takes	Typical access	Typical maintenance
1 — Publish or update served files	Add or edit files the site already knows how to serve (`robots.txt`, `sitemap.xml`, a data dictionary, a mapping table).	Permission to write at a well-known path, or an administrator who can place the file. A CDN or hosted platform can override what you upload.	Keep the file current when paths, policy, or inventory change.
2 — Configure the publishing platform	Change HTTP headers, HTML templates, landing-page markup, content negotiation, CDN or firewall rules, or repository-platform settings.	Platform, web, or network administrator access; sometimes a developer.	Recheck after platform, theme, or CDN changes.
3 — Write or run software	Implement or operate an API or MCP server, or run specialized tooling such as a validator.	Developer time and, if you host a service, operational support.	Code, dependencies, and (only if you host it) the running service.

A Level 1 file can still require administrator access to publish. A Level 3 check such as SHACL validation can run **locally** or in a batch pipeline; it does not require a public validation service unless you choose to offer one.

Optional and emerging items are labeled. Skipping them is not a failure.

Three different kinds of statement

Do not collapse these into one “AI policy” switch.

1. **Crawl preferences** — `robots.txt` (RFC 9309) tells cooperating clients what they may fetch. Compliance is voluntary. A preference is not a license.
2. **Access enforcement** — HTTP 401/403, CDN or WAF rules, rate limits (`429` + `Retry-After`), and authentication actually allow or deny a request.
3. **Rights and usage signals** — licenses, TDM reservation declarations, and draft usage vocabularies state permissions or preferences about *use*. They do not, by themselves, block a fetch.

US and EU practice around text-and-data-mining reservations is not the same. Read TDM reservations before treating any signal as a legal conclusion.

How to use this checklist

- Work through the four webinar facets: **Reachable**, **Readable**, **Callable**, and **Meaningful**. Evaluation sits beside them; it is not a fifth grade.
- Each item names the action, who can typically do it, how to verify it, and where the webinar or a handout discusses it.
- Print the PDF and mark the empty boxes, or copy this Markdown file and track status in your own notes. The generated PDF is printable; it is not an interactive fill-in form.
- Generate a Level 1 view or a Levels 1–2 view from this same file if you want a shorter handout (see README).
- MCP is **not** required in order to act on the webinar. Reachable and Readable work plus a well documented API already go a long way.

Reachable

Network-accessible on terms you have thought through: signal what you expect, enforce the constraints that matter, and look at logs. See webinar PDF pp. 10–13 and the glossary section Being reachable.

☐ RCH-02. Publish a robots.txt at the site root

Level 1 · Reachable · preference signal · established

ACCESS NEEDED	Permission to place a file at the origin site root, or an administrator who can. A CDN-managed `robots.txt` can replace the file you uploaded.
MAINTENANCE	Review when paths, collections, or vendor product tokens change.
SOURCES	Webinar PDF p. 11; Glossary — robots.txt; Example robots.txt; RFC 9309

DO THIS

Publish `https://YOUR-HOST/robots.txt`. Use `User-agent` groups with `Allow` and `Disallow` path prefixes that begin with `/` (for example `Disallow: /*.gif$`, not a bare `*.gif$`). Remember that this file states preferences for cooperating clients; it cannot make a client comply, and it does not grant or deny copyright permission.

An annotated example lives with the webinar assets. Treat it as a demonstration (replace hosts and placeholder paths) rather than a drop-in policy.

HOW TO VERIFY

`GET https://YOUR-HOST/robots.txt` and confirm the bytes you expect are served (not a generic platform default). Check that every `Allow`/`Disallow` path starts with `/`.

> **WATCH FOR**
>
> RFC 9309 does not define `Crawl-delay`. Prefer HTTP `429` and `Retry-After` for rate control (see RCH-04). Some clients ignore the file entirely.

☐ RCH-03. Separate trainer, index, and user-triggered visitors

Level 1 · Reachable · preference signal · established

ACCESS NEEDED	Same as publishing `robots.txt`, plus someone who can reconcile that file with CDN rules.
MAINTENANCE	Recheck vendor bot names and documented behavior; they change.
SOURCES	Webinar PDF p. 12; Glossary — trainer crawler, index builder, fetcher; Identifying agents; Google-Extended

DO THIS

Do not rely on a single `User-agent: *` rule if you want different outcomes for different missions. The webinar groups visitors roughly as:

1. **Trainer crawlers** (examples named on the slide include GPTBot, ClaudeBot, CCBot, and Google-Extended) collecting content for model training or related use.
2. **Search or index crawlers** (examples include OAI-SearchBot, PerplexityBot, and Googlebot for AI Overviews) that can affect discovery and citation.
3. **User-triggered fetchers** (examples include ChatGPT-User and Perplexity-User) retrieving a page because a person asked. Several vendors document that these often do **not** honor `robots.txt`.
4. **Unknown** clients, where a robots line is only a guess and enforcement may belong at the firewall.

Write named groups for the products you actually care about, and keep CDN rules consistent with that intent.

If you mention **Google-Extended**, treat it as a *product token* with no separate HTTP User-Agent. It covers Gemini **training and grounding**. It does **not** affect inclusion or ranking in Google Search. Do not say that blocking it leaves all “AI search visibility” unchanged.

HOW TO VERIFY

Read `robots.txt` and confirm that trainer tokens and search tokens are not forced into one outcome unless that is a deliberate choice. Compare each named token with the vendor’s current crawler documentation. For Google-Extended, use Google’s crawler documentation rather than assuming it behaves like Googlebot.

> **WATCH FOR**
>
> The PDF face still contains the broader sentence “Blocking does not affect AI search visibility.” Prefer the glossary and Google’s documentation, as above.

☐ RCH-06. Optionally declare IETF AI usage preferences

Level 1 · Reachable · rights signal · draft · Optional

ACCESS NEEDED	Same as editing `robots.txt` for a `Content-Usage` line; HTTP header attachment needs platform or CDN configuration (Level 2).
MAINTENANCE	Revisit when the Internet-Drafts are revised or published as RFCs.
SOURCES	Webinar PDF p. 11; Glossary — AI Preferences (AIPREF); Bibliography — Being reachable; AIPREF vocabulary draft-08; AIPREF attachment draft-05; TDM reservations

DO THIS

If you want a machine-readable *usage preference* in addition to crawl rules, you may add IETF AI Preferences syntax—for example `Content-Usage: search=y, train-ai=n`—in `robots.txt` and/or as an HTTP `Content-Usage` header. Label this as an **IETF Internet-Draft** mechanism (`draft-ietf-aipref-vocab-08` and `draft-ietf-aipref-attach-05`), not a finished RFC.

Cloudflare Content Signals (`Content-Signal`, with labels such as `search`, `ai-input`, and `ai-train`) are a **different** vocabulary from AIPREF and are also distinct from Cloudflare's default AI-bot blocking controls. If you use both lines, keep them separate; a parser for one will not read the other.

HOW TO VERIFY

If you adopted the robots-file form, open `/robots.txt` and confirm the `Content-Usage` line matches the drafts you intended. If you adopted the header form, request a sample page and inspect the `Content-Usage` response header. Confirm you did not accidentally treat the line as a fetch block.

> **WATCH FOR**
>
> The webinar PDF shows `Content-Usage` without calling it a draft. The handouts use vocabulary **-08**; the example `assets/robots.txt` comments still mention **-07** in places—follow the glossary and bibliography.

☐ RCH-07. Optionally point at an RSL license document with an absolute URL

Level 1 · Reachable · rights signal · emerging · Optional

ACCESS NEEDED	Permission to publish an RSL XML document and to edit `robots.txt`.
MAINTENANCE	Keep the XML and the `License:` URL in agreement if licensing changes.
SOURCES	Webinar PDF p. 11; Glossary — Really Simple Licensing; TDM reservations; RSL specification

DO THIS

Really Simple Licensing is an emerging XML licensing vocabulary. If you use a `License:` line in `robots.txt`, the value must be an **absolute URL** to that XML document, such as `https://example.org/rsl.xml`, not a host-relative path such as `/rsl.xml`.

HOW TO VERIFY

Confirm the `License:` value starts with `https://` (or another absolute scheme), fetch it, and confirm it returns the RSL document you intended.

WATCH FOR

The PDF face still shows `License: /rsl.xml`. Prefer the glossary, bibliography, and RSL specification. RSL is earlier-stage than several other rights tools discussed in the webinar; skip it unless you have a reason to experiment.

Readable

Expressive resources: inventories, landing pages, and typed links that an agent can follow without brittle scraping. See webinar PDF pp. 14–20 and Being readable.

☐ RD-01. Publish a sitemap and advertise it with absolute, file-wide Sitemap lines

Level 1 · Readable · standard practice · established

ACCESS NEEDED	Permission to publish XML at a stable URL and to edit `robots.txt`.
MAINTENANCE	Update `lastmod` when pages actually change; stale sitemaps are common, and crawlers may walk the site anyway.
SOURCES	Webinar PDF p. 14; Glossary — Sitemap; Example sitemap.xml; sitemaps.org protocol; Google robots.txt sitemap field

DO THIS

Publish a sitemap (or several thematic sitemaps, optionally listed from a sitemap index) using the sitemaps.org vocabulary. List each sitemap with an absolute `Sitemap:` URL in `robots.txt`. Those declarations are **not** scoped to `User-agent` groups; they apply file-wide. You can publish separate sitemaps for different parts of the site, but placing a `Sitemap:` line under a particular User-agent block does not make it bot-specific.

Omit URLs that `robots.txt` disallows. If you include `lastmod`, use the date the page changed, not the date the sitemap file was regenerated. A sitemap is an inventory, not crawl permission and not a reading guide (that role is closer to an optional `llms.txt`).

HOW TO VERIFY

Fetch `/robots.txt` and each advertised sitemap URL. Confirm the XML is well formed, each `loc` is on your host, a sample of those URLs respond, and disallowed paths are absent. Confirm `Sitemap:` lines use absolute URLs.

WATCH FOR

The PDF face says you can “express them for specific user agents.” Prefer the glossary, bibliography, and Google’s robots.txt documentation: the sitemap field “isn’t tied to any specific user agent.”

☐ RD-05. Optionally publish llms.txt

Level 1 · Readable · advice · emerging · Optional

ACCESS NEEDED	Permission to place `llms.txt` at the site root or in a covering subfolder.
MAINTENANCE	Curate the reading list; an outdated guide can be worse than none.
SOURCES	Webinar PDF p. 15; Glossary — llms.txt; Example llms.txt; llms.txt; Chrome Lighthouse llms.txt audit

DO THIS

`llms.txt` is an optional, curated reading order for agents, not a baseline requirement for following the webinar. Version 2 (dated August 2026 in the deck) also describes pointing from each page to a Markdown alternate (`rel="alternate"` type `text/markdown`) and to the covering `llms.txt` (`rel="describedby"`), as HTML tags or HTTP `Link` headers. A file in a subfolder can cover the pages under it.

Clean semantic HTML, Signposting, and Markdown negotiation can address similar goals without this file.

HOW TO VERIFY

`GET https://YOUR-HOST/llms.txt` (or the subfolder URL) and confirm the links you listed are the ones you intended. If you implemented v2 page pointers, inspect `rel="alternate"` and `rel="describedby"` on a sample page. An optional Chrome Lighthouse agentic-browsing audit exists for this arrangement.

> **WATCH FOR**
>
> Adoption is still limited. Bibliography notes attribute roughly 10% penetration to an SE Ranking analysis of nearly 300,000 domains (about 10.13% in that study)—not “10% of the web.” Do not treat the file as required for SI-oriented access.

☐ RD-06. Optionally document a workflow as SKILL.md

Level 1 · Readable · advice · emerging · Optional

ACCESS NEEDED	Permission to publish a Markdown file or folder; adding executable scripts or service calls raises this to Level 3.
MAINTENANCE	Update the skill when the API or workflow changes.
SOURCES	Webinar PDF p. 18; Glossary — Agent skill; Agent Skills

DO THIS

If you want to tell an agent how to perform a specific job (for example, how to call your records API), you may publish a `SKILL.md` with YAML front matter and imperative instructions. The webinar treats this as documenting a workflow. A Markdown-only skill is a static file. Packaging scripts, tool calls, or live service invocations is optional and needs software maintenance.

HOW TO VERIFY

Fetch the `SKILL.md` URL. Confirm it states when to use the skill, points at the real documentation or OpenAPI file, and does not contradict those sources.

☐ RD-07. Optionally experiment with ard.json

Level 1 · Readable · advice · bleeding-edge · Optional

ACCESS NEEDED	Permission to publish `/.well-known/ard.json`.
MAINTENANCE	High churn; specifications in this area change quickly.
SOURCES	Webinar PDF p. 19; Glossary — ARD / ard.json; Example ard.json; ARD specification

DO THIS

Only if you are experimenting: publish an Agentic Resource Discovery file at `/.well-known/ard.json`. The webinar marks this as bleeding edge. It is not a substitute for `robots.txt`, a sitemap, Signposting, or a documented API.

HOW TO VERIFY

Fetch `https://YOUR-HOST/.well-known/ard.json` and confirm it parses as JSON and matches the current specification you intended to follow.

Meaningful

Structured context—identifiers, vocabularies, profiles, rights, provenance, and optional shapes or mappings—so a model is less likely to invent your schema. See webinar PDF pp. 27–34 and Being meaningful. A data dictionary is already useful; a formal ontology is not automatically better.

☐ MEA-02. Publish a data dictionary or vocabulary at a stable URL

Level 1 · Meaningful · advice · established

ACCESS NEEDED	Permission to publish a page or file at a stable URL. Building a formal ontology is a larger modeling effort, not a higher "grade."
MAINTENANCE	Version the vocabulary when terms change.
SOURCES	Webinar PDF p. 29; Glossary — vocabulary, data dictionary; From terms to ontologies; Vocabulary tools

DO THIS

Publish the words and fields you use: at minimum an informal data dictionary with definitions, units, and allowed values. A term list, a taxonomy, or an ontology adds structure for some use cases; choose a proportionate approach rather than climbing a ladder for its own sake. Reuse an existing vocabulary when one already covers the subject.

HOW TO VERIFY

Fetch the stable URL. Confirm a new staff member (or an agent) can look up a core field such as an assay or study type and find a definition.

☐ MEA-07. Optionally publish a TDM reservation signal

Level 1 · Meaningful · rights signal · emerging · Optional

ACCESS NEEDED	Permission to publish `/.well-known/tdmrep.json` and/or to set HTTP headers or HTML metadata (headers are Level 2).
MAINTENANCE	Keep the site-level file and per-resource declarations consistent.
SOURCES	Webinar PDF p. 31; TDM reservations; TDMRep CG Final Report; Bibliography — Rights, credit, and lineage

DO THIS

TDMRep is a W3C Community Group Final Report (not a W3C Recommendation) for machine-readable text-and-data-mining rights signals, including `/.well-known/tdmrep.json`, HTTP headers, and HTML metadata. A `tdm-reservation` value of `1` reserves TDM rights in that declaration; `0` means they are not reserved by that declaration. An optional `tdm-policy` URL can point to more detail.

This is a **rights signal**, not a crawler block, and not a substitute for reading applicable law or licenses. It is commonly discussed in EU and STM reservation settings; US practice differs. This item is not legal advice.

HOW TO VERIFY

If you publish the well-known file, fetch it and confirm `tdm-reservation` (and `tdm-policy`, if used) match your intended statement. Do not expect the file alone to change whether a client can GET a URL.

☐ MEA-10. Optionally publish mappings as SSSOM/TSV

Level 1 · Meaningful · standard practice · established · Optional

ACCESS NEEDED	Permission to publish a file, plus people who understand both vocabularies well enough to map them. Creating high-quality mappings is skilled work; hosting a mapping *service* would be Level 3.
MAINTENANCE	Update mappings when either vocabulary versions.
SOURCES	Webinar PDF p. 34; Glossary — SSSOM; SSSOM; SSSOM specification (dev)

DO THIS

If you need to say “our term X matches term Y,” publish a mapping set using SSSOM. The common exchange format is **SSSOM/TSV** (not CSV) with mapping-set metadata. Each row records the two terms and a relation such as exact, close, broader, or related, and it can record a justification. Fields such as `author_id` and `confidence` are **optional**; do not imply that every row names an author or carries an independently established confidence value.

HOW TO VERIFY

Fetch the TSV (or the URL you advertised). Confirm it is tab-separated, includes the mapping columns you intended, and that optional provenance columns are present only when you actually have the information.

Evaluation

Evaluation is a large topic; the webinar treats it as a place to start, not a scoring system. Different checks answer different questions. See webinar PDF p. 36, Checking the result, and Evaluation.

☐ EVL-01. Try an agent against your public documentation and record what happens

Level 1 · Evaluation · advice · established

ACCESS NEEDED	Anyone who can use a public chat or agent interface and read your public pages.
MAINTENANCE	Repeat after you change robots rules, sitemaps, or landing pages.
SOURCES	Webinar PDF p. 36; Evaluation prompts (handout); prompts/evals

DO THIS

Give an agent or a simple chat UI links to your site or API documentation and see what it can find. Write down the question, whether the answer was supported by fetched evidence, and what was missing (license, DOI, data dictionary, and so on). Starting prompts live in this repository under `prompts/evals`.

HOW TO VERIFY

Keep a short written record (even a few rows) of questions, dates, tools used, and whether the supporting URL was correct.

☐ EVL-02. Keep access checks, answerability tests, and accuracy reviews separate

Level 1 · Evaluation · advice · established

ACCESS NEEDED	The people who run documentation and metadata review.
MAINTENANCE	Do not replace human review of content with a single vendor “agent score.”
SOURCES	Evaluation; Webinar PDF p. 36

DO THIS

Ask three different questions and do not merge the answers into one grade:

1. Can an agent *discover and read* the documentation (access and readability)?
2. Can it *answer the questions your users care about*, with evidence?
3. Are the docs and examples *accurate*?

Tools mentioned in the evaluation handout (including early-stage documentation checkers) may help with (1). Competency questions help with (2). Human review, and tools that execute documented examples, help with (3). No single score covers all three, and a passing readability check is not a measure of trustworthiness.

HOW TO VERIFY

In your notes, label each test as access/readability, answerability, or accuracy. If you record a numeric score from a tool, record what that tool actually measured.

☐ EVL-03. Optionally reuse published audits and portal comparisons

Level 1 · Evaluation · advice · emerging · Optional

ACCESS NEEDED	A browser, and (for NDE) a public portal account if the feature requires one.
MAINTENANCE	Repeat when you change `llms.txt` or portal metadata.
SOURCES	Webinar PDF p. 36; NDE AI-assisted search; Chrome Lighthouse llms.txt audit

DO THIS

If they apply to you: compare NIAID Data Ecosystem search with AI-assisted search on and off; run Chrome Lighthouse’s `llms.txt` audit if you publish that file; try the webinar evaluation prompts. These are diagnostics, not badges.

HOW TO VERIFY

Save the date and the tool output (or a screenshot) next to the checklist item you thought you had completed.

Closing reminder

Other than APIs and MCP services, almost everything in the webinar is a static text file your server can expose. That does not make the work easy: the files need governance and curation. Prefer durable investments that help human users as well as agents. See webinar PDF p. 37 and the resource list on p. 38.

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